PRACTICAL 1:

Workbook Basics

AIM:

To create, save, rename, delete, move and copy worksheets in MS Excel.

DESCRIPTION:

This practical explains complete workbook handling in MS Excel. It includes creating a new workbook, saving it properly, and managing worksheets efficiently.

STEPS:

1. Opening Microsoft Excel

   i. Open your computer and click on the **Start Menu**.

   ii. Type *Microsoft Excel* in the search bar and press **Enter**.

   iii. Wait for the Excel home screen to appear.

2. Creating a New Workbook

   i. On the home screen, click on **Blank Workbook**.

   ii. A new workbook will open with a default sheet named *Sheet1*.

3. Saving the Workbook

   i. Click on the **File** tab in the top left corner.

   ii. Select **Save As**. Choose the location (Desktop/Documents).

   iii. In the File Name box, type *Practical1_Workbook*. Click **Save**.

   iv. This stores your workbook safely.

4. Renaming a Worksheet

   i. At the bottom of the Excel window, locate *Sheet1*.

   ii. Right-click on it and select **Rename**. Type the new name (Example: StudentData) and press **Enter**.

   iii. The sheet name will change.

5. Inserting a New Worksheet

Click on the **+ (Plus icon)** next to the sheet tabs. A new worksheet will be inserted automatically.

6. Deleting a Worksheet

Right-click on the sheet you want to remove. Click **Delete** and confirm if Excel asks for confirmation.

7. Moving a Worksheet

i. Right-click on the sheet tab. Select **Move or Copy**.

ii. Choose the position where you want the sheet to appear. Click **OK**.

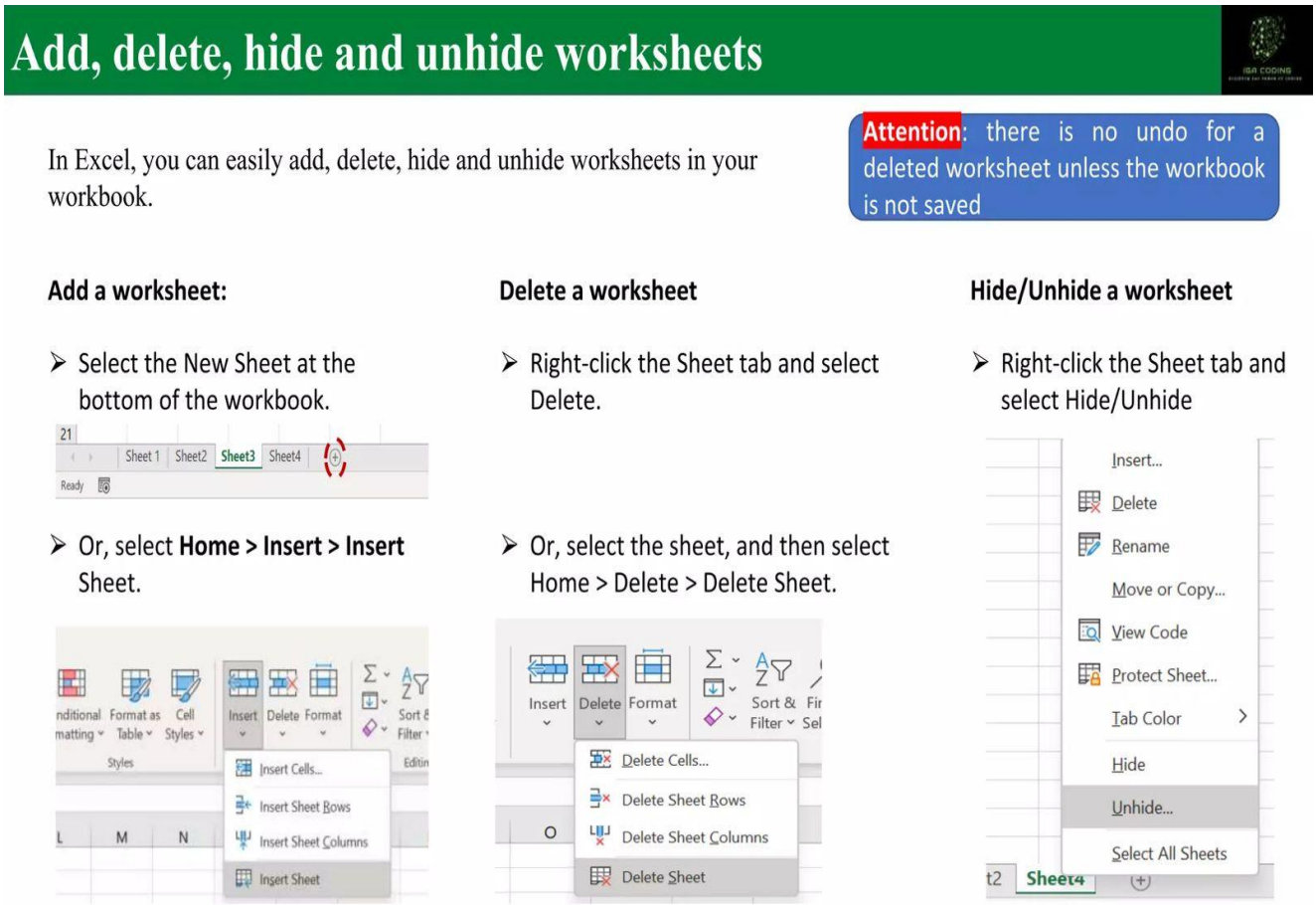
8. Copying a Worksheet

Right-click on the sheet tab again. Choose **Move or Copy**. Tick the option **Create a Copy**. Click **OK**. A duplicate sheet will be created.

RESULT:

The workbook was successfully created, saved, and worksheets were renamed, inserted, deleted, moved and copied.

OUTPUT IMAGE:



PRACTICAL 2:

Data Entry and Editing

AIM:

To enter employee data and perform calculations using formulas.

DESCRIPTION:

This practical explains structured data entry and use of basic Excel formulas like SUM, AVERAGE and MAX.

STEPS:

1. Creating Table Headings

   i. Open a new worksheet.

   ii. In cell A1 type **Employee Name**, in B1 type **Department**, and in C1 type **Salary**.

   iii. These headings define the table structure.

2. Entering Employee Records

   i. Start entering employee names from A2 to A11.

   ii. Enter departments in B2 to B11 and salary values in C2 to C11.

3. Editing Incorrect Data

   i. If you make a mistake, double-click on the cell containing the error.

   ii. Correct the value and press **Enter**.

   iii. Excel will update the information without deleting the entire table.

4. Calculating Total Salary

   i. Click on cell C12. Type the formula =SUM(C2:C11) and press **Enter**.

   ii. Excel will calculate the total salary of all employees.

5. Calculating Average Salary

   i. Click on cell C13. Type =AVERAGE(C2:C11) and press **Enter**.

   ii. The average salary will be displayed.

6. Finding Maximum Salary

Click on cell C14. Type =MAX(C2:C11) and press **Enter**. The highest salary value will appear.

RESULT:

Employee data was entered and total, average and maximum salary were calculated successfully.

OUTPUT IMAGE:

	A	B	C	D
1	**Employee Name**	**Department**	**Salary (₹)**	
2	Rahul Sharma	HR	35,000	
3	Priya Patel	Finance	42,000	
4	Amit Verma	IT	50,000	
5	Neha Singh	Marketing	38,000	
6	Arjun Mehta	Sales	36,000	
7	Kavita Nair	Administration	34,000	
8	Rohan Gupta	IT	48,000	
9	Sneha Iyer	Finance	41,000	
10	Vikram Shah	Marketing	39,000	
11	Pooja Desai	HR	37,000	
12		Total Salary	400,000	
13		Average Salary	40,000	
14		Maximum Salary	50,000	
15				

PRACTICAL 3:

Filling Series and Drag Fill

AIM:

To generate number series and copy formulas using AutoFill feature.

DESCRIPTION:

Excel allows automatic generation of number sequences and quick copying of formulas using the Fill Handle.

STEPS:

1. Generating Number Series

   i. Click on cell A1 and type 1.

   ii. Go to **Home → Fill → Series**.

   iii. Choose Columns, set Step Value as 1 and Stop Value as 50. Click **OK**.

   iv. Numbers 1 to 50 will be generated automatically.

2. Applying a Formula

   i. In cell B1 type =A1*2 and press Enter.

   ii. This multiplies the value of A1 by 2.

3. Using Drag Fill

   i. Move the cursor to the bottom-right corner of B1 until the small black cross appears.

   ii. Click and drag down to B50. Release the mouse.

   iii. The formula will automatically copy and adjust for each row.

RESULT:

Serial numbers and formulas were generated successfully using Fill Series and Drag Fill.

OUTPUT IMAGE:

	A	B
1	1	
2	2	
3	3	
4	4	
5	5	
6	6	
7	7	
8	8	
9	9	
10	10	
11	11	
12	12	
13	13	
14	14	
15	15	
16	16	
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36	36	
37	37	
38	38	
39	39	
40	40	
41	41	
42	42	
43	43	
44	44	
45	45	
46	46	
47	47	
48	48	
49	49	
50	50	
51		

PRACTICAL 4:

Data Entry and Formatting

AIM:

To prepare and format a student mark sheet.

DESCRIPTION:

Formatting improves the presentation of data using borders, bold text, alignment and AutoFit options.

STEPS:

1. Creating the Mark Sheet Structure

Enter headings: Name, Roll No, Subject1, Subject2, Subject3, Total in the first row.

2. Entering Student Marks

Enter data for 10 students below the headings.

3. Calculating Total Marks

In the Total column, type =SUM(C2:E2) and press Enter. Drag the formula down for all students.

4. Applying Borders

Select the entire table. Go to Home → Borders → All Borders to create visible table lines.

5. Formatting Headings

Select the heading row and click **Bold**. You may also apply center alignment.

6. Adjusting Column Width

Select all columns and double-click between column headers to apply AutoFit.

RESULT:

Student mark sheet was created and formatted properly.

OUTPUT IMAGE:

	A	B	C	D	E	F	G
1	Name	Roll No	Maths	Science	English	Total	
2	Rahul Sharma	101	78	82	75	235	
3	Priya Patel	102	85	79	80	244	
4	Amit Verma	103	90	88	84	262	
5	Neha Singh	104	72	76	70	218	
6	Arjun Mehta	105	68	74	72	214	
7	Kavita Nair	106	88	85	90	263	
8	Rohan Gupta	107	76	80	78	234	
9	Sneha Iyer	108	82	86	84	252	
10	Vikram Shah	109	70	75	73	218	
11	Pooja Desai	110	92	89	91	272	
12							

PRACTICAL 5:

Sorting and Filtering

AIM:

To sort and filter data using single and multiple criteria.

DESCRIPTION:

Sorting helps arrange data in ascending or descending order, while filtering displays only selected records based on given conditions.

STEPS:

1. Preparing the Dataset

   i. Open the worksheet containing employee or student data.

   ii. Ensure that the table has proper headings and no empty rows or columns in between.

2. Sorting Data in Descending Order

   i. Select the complete dataset including headings.

   ii. Go to the **Data** tab on the ribbon. Click on **Sort**.

   iii. In the Sort dialog box, choose the column (for example, Total Marks or Salary).

   iv. Under Order, select **Largest to Smallest**. Click **OK**. The data will now be arranged in descending order.

3. Applying Filter to Dataset

   i. Select the dataset again. Go to the **Data** tab and click on **Filter**.

   ii. Small drop-down arrows will appear beside each column heading.

4. Filtering by Single Criteria

   i. Click on the drop-down arrow of the Department column.

   ii. Uncheck Select All and choose only **HR**. Click **OK**.

Only employees from HR department will be displayed.

5. Applying Multiple Criteria Filter

Click on the Salary column drop-down.

Select **Number Filters → Greater Than**. Enter 40000 and click OK.

Now apply another filter in Department column and select **Sales**. Only records satisfying both conditions will appear.

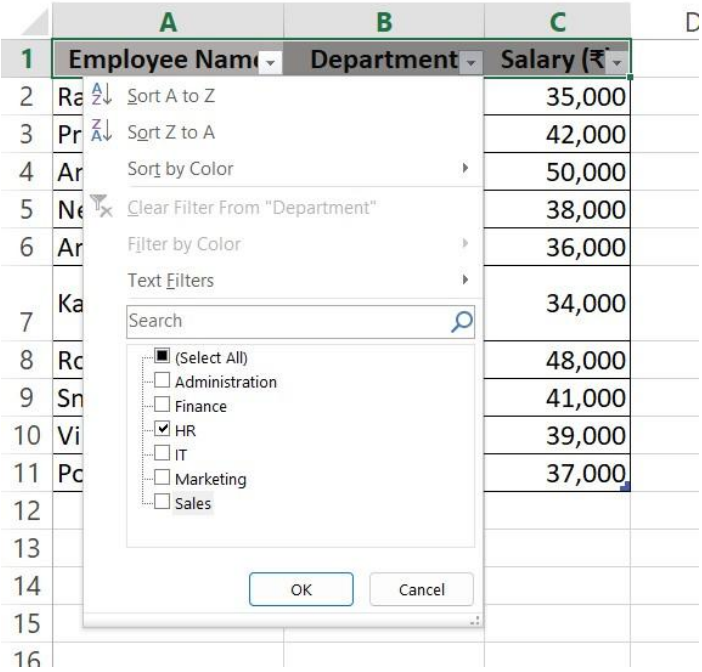
RESULT:

Data was successfully sorted and filtered using single and multiple conditions.

OUTPUT IMAGE:

Name	Roll No	Maths	Science	English	Total
Pooja Desai	110	92	89	91	272
Kavita Nair	106	88	85	90	263
Amit Verma	103	90	88	84	262
Sneha Iyer	108	82	86	84	252
Priya Patel	102	85	79	80	244
Rahul Sharma	101	78	82	75	235
Rohan Gupta	107	76	80	78	234
Neha Singh	104	72	76	70	218
Vikram Shah	109	70	75	73	218
Arjun Mehta	105	68	74	72	214

Employee dataset to show only employees in the HR department



Employee Name	Department	Salary (₹)
Rahul Sharma	HR	35,000
Pooja Desai	HR	37,000

The use of multiple criteria in filtering

	A	B	C	D
1	Employee Name	Department	Salary (₹	
4	Amit Verma	Sales	50,000	
6	Arjun Mehta	Sales	42,000	
12				

PRACTICAL 6:

Working with Charts

AIM:

To create Bar Chart, Line Chart and Combo Chart in MS Excel.

DESCRIPTION:

Charts are graphical representations of data that make comparison and analysis easier.

STEPS:

1. Creating a Bar Chart

   i. Select the student marks dataset including headings.

   ii. Go to the **Insert** tab. In the Charts group, click on **Bar Chart** and choose a suitable 2D Bar Chart.

   iii. The chart will be inserted into the worksheet.

2. Adding Chart Elements

   i. Click on the chart. Use the **Chart Design** tab to add Chart Title and Axis Titles.

   ii. Edit the chart title by clicking on it and typing a suitable name.

3. Creating a Line Chart

   i. Select monthly sales data. Go to **Insert → Line Chart**.

   ii. Choose a 2D Line Chart. The line chart will display sales trend over months.

4. Creating a Combo Chart

   i. Select revenue and profit data.

   ii. Go to **Insert → Combo Chart**. Choose Custom Combo Chart.

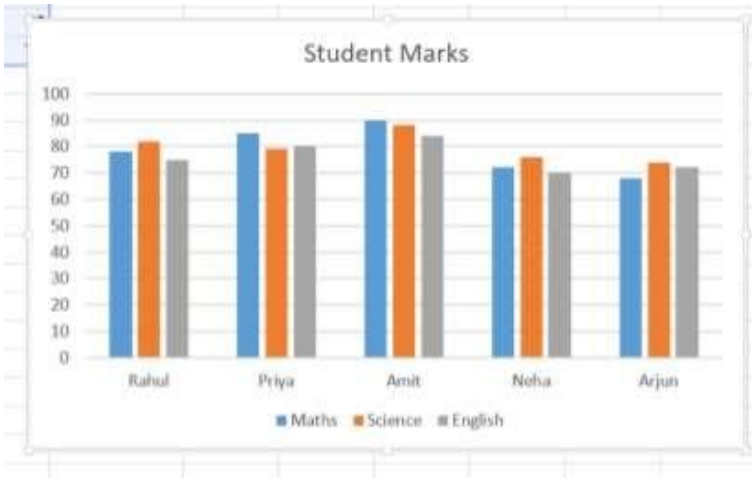
   iii. Set Revenue as Column Chart and Profit as Line Chart. Click OK.

RESULT:

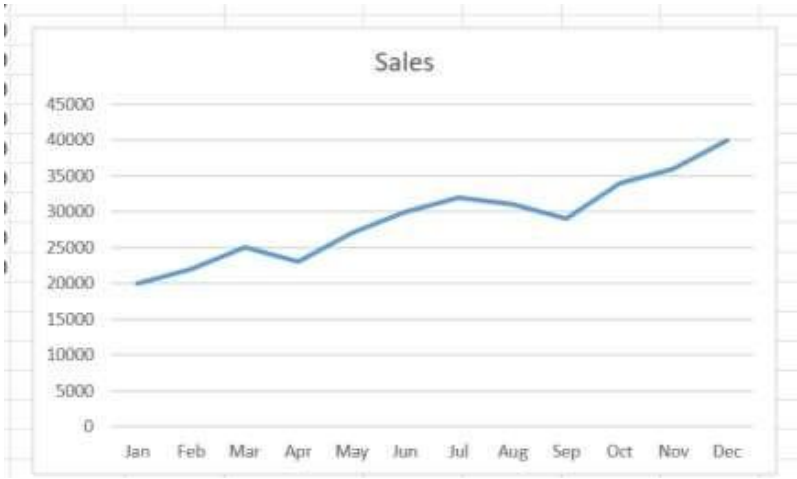
Bar chart, line chart and combo chart were created successfully to represent data visually.

OUTPUT IMAGE:

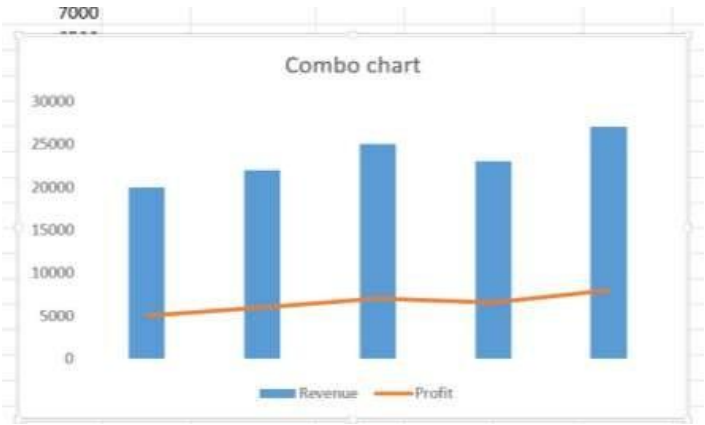
BAR CHART :



LINE CHART :



COMBO CHART:



PRACTICAL 7:

Advanced Data Analysis

AIM:

To use Goal Seek, What-If Analysis and Scenario Manager.

DESCRIPTION:

Advanced tools in Excel help in forecasting and decision-making by analyzing different data possibilities.

STEPS:

1. Using Goal Seek

   i. Prepare a worksheet with marks and average formula.

   ii. Go to **Data → What-If Analysis → Goal Seek**.

   iii. In the dialog box, set the cell containing average marks as 'Set Cell'.

   iv. Enter desired average value in 'To Value'. In 'By Changing Cell', select the last exam marks cell.

   v. Click OK. Excel will calculate the required marks.

2. Using What-If Analysis

   i. Create a salary increment sheet. Change increment percentage manually and observe how total salary changes.

   ii. This helps analyze different possible outcomes.

3. Using Scenario Manager

   i. Go to **Data → What-If Analysis → Scenario Manager**. Click Add.

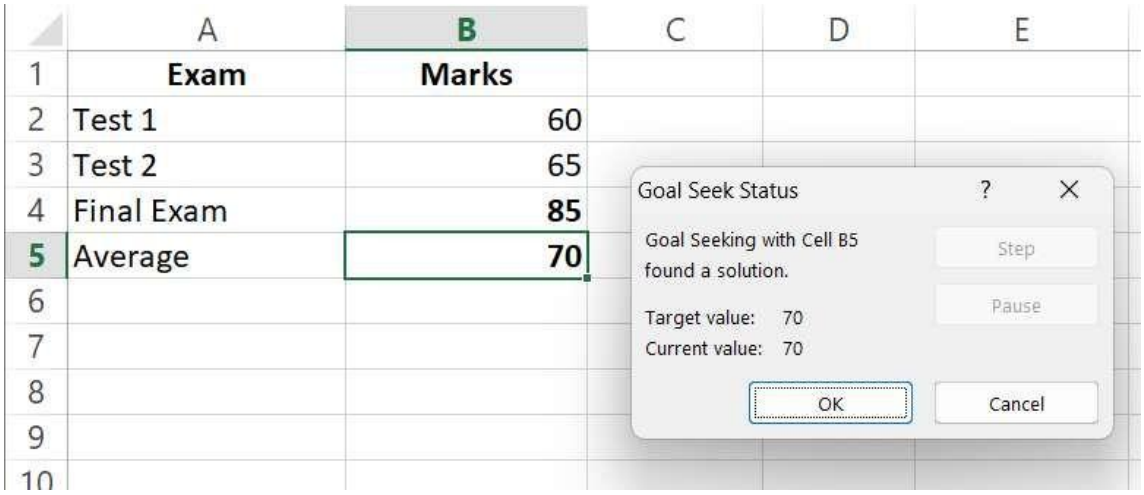
   ii. Name the scenario (Example: High Investment). Select changing cells.

   iii. Enter new values and click OK. Add multiple scenarios and use Summary option to compare results.

RESULT:

Advanced tools were successfully used to analyze different data situations.

OUTPUT IMAGE:



	A	B	C	D	E
1	**Exam**	**Marks**			
2	Test 1	60			
3	Test 2	65			
4	Final Exam	**85**			
5	Average	**70**			
6					
7					
8					
9					
10					

PRACTICAL 8:

Text Manipulation

AIM:

To use text functions like CONCAT, LEFT, RIGHT, MID and LEN.

DESCRIPTION:

Text functions help combine, extract and analyze text data efficiently.

STEPS:

1. Using CONCAT Function

Enter First Name in column A and Last Name in column B. In column C type =CONCAT(A2,” “,B2) and press Enter. The full name will appear.

2. Using LEFT Function

In a new cell type =LEFT(A2,3) and press Enter. The first three characters will be extracted.

3. Using RIGHT Function

Type =RIGHT(A2,2) and press Enter. The last two characters will be extracted.

4. Using MID Function

Type =MID(A2,2,4) and press Enter. This extracts characters starting from position 2 up to 4 characters.

5. Using LEN Function

Type =LEN(A2) and press Enter. Excel will count total number of characters in the text.

RESULT:

Text functions were applied successfully to combine and extract text.

OUTPUT IMAGE:

	A	B	C	D
1	Current Salary	Increment %	New Salary	
2	40000	5%	42000	
3	40000	10%	44000	
4	40000	15%	46000	
5				

PRACTICAL 9:

Using Add-ins in MS Excel

AIM:

To install and use Data Analysis Toolpak Add-in.

DESCRIPTION:

Add-ins increase Excel's functionality by adding advanced data analysis tools.

STEPS:

1. Opening Add-ins Window

Click on **File → Options**. Select **Add-ins** from the left panel.

2. Installing Data Analysis Toolpak

At the bottom, select **Excel Add-ins** and click **Go**. Tick **Analysis ToolPak** and click OK.

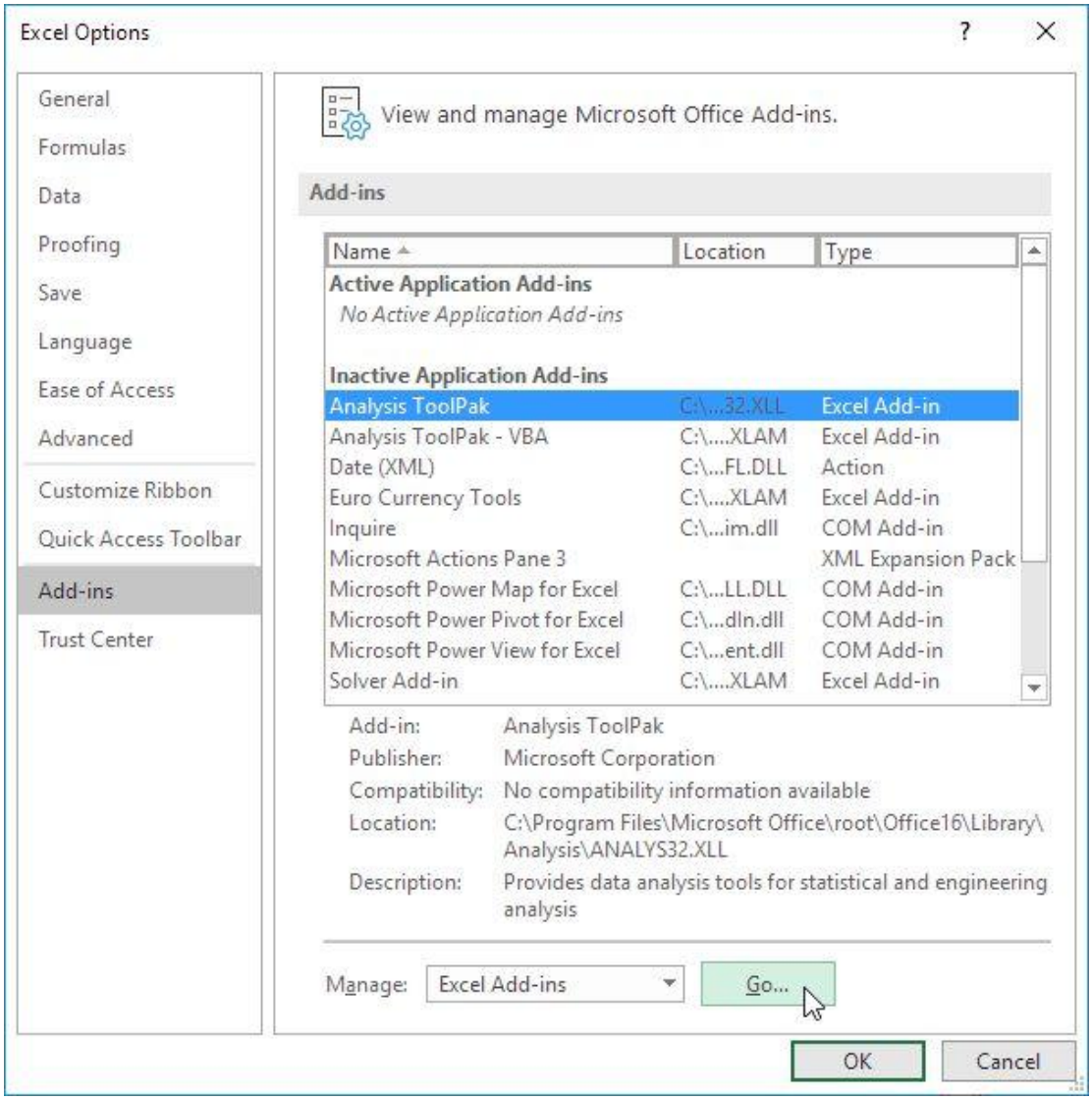
3. Using Data Analysis Tool

Go to **Data tab**. Click on **Data Analysis**. Choose any tool like Descriptive Statistics. Select input range and click OK.

RESULT:

Data Analysis Toolpak was installed and used successfully.

OUTPUT IMAGE:



PRACTICAL 10:

SharePoint and Add-ins Integration

AIM:

To understand SharePoint integration with Excel for collaboration.

DESCRIPTION:

SharePoint allows multiple users to access and collaborate on Excel files online.

STEPS:

1. Opening SharePoint Site

Open a web browser and log in to your SharePoint account.

2. Uploading Excel File

Go to **Document Library**. Click on **Upload** and select the Excel file from your computer.

3. Sharing the File

Click on the uploaded file. Select **Share** option. Enter email addresses of team members and click Send.

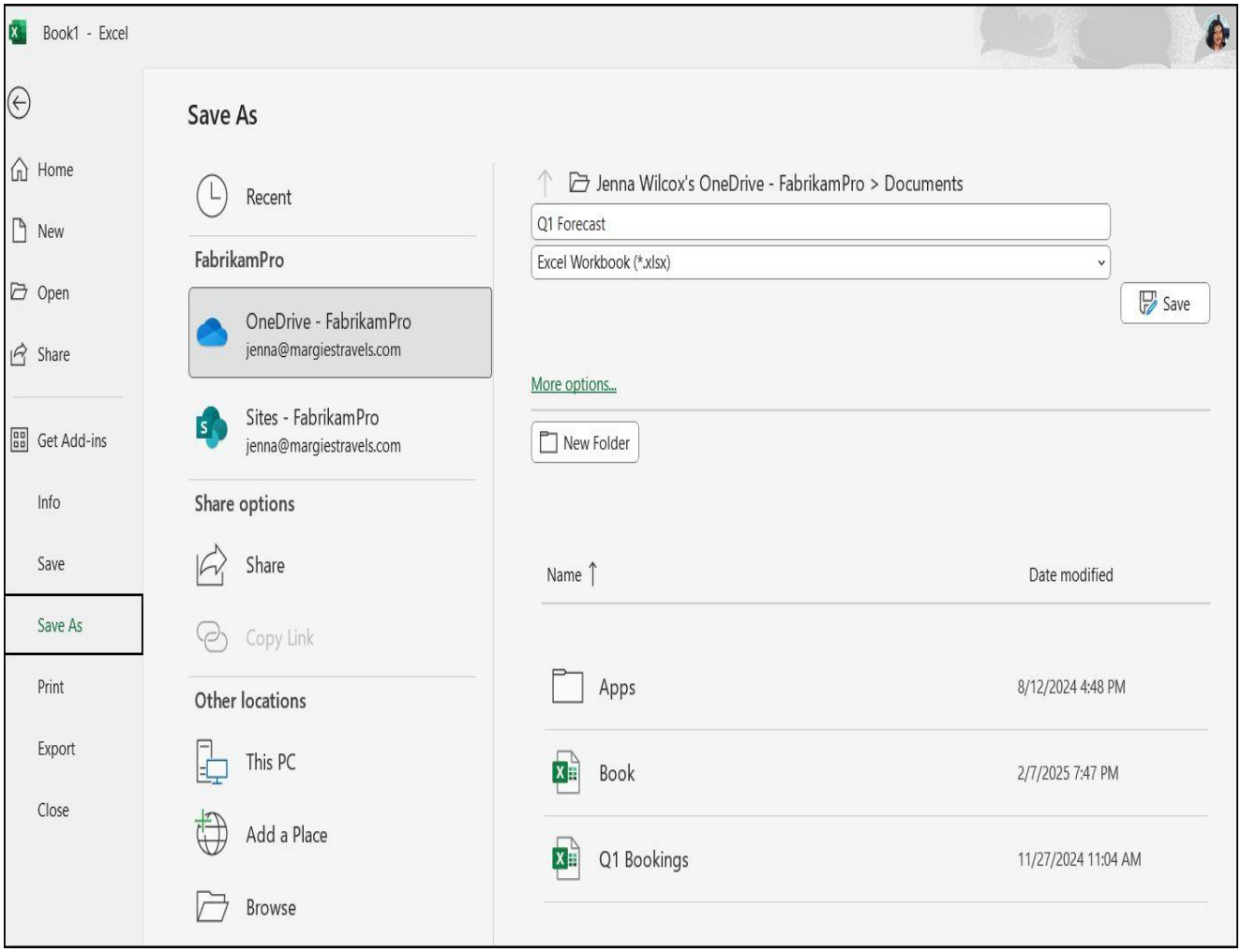
4. Collaborating Online

Team members can open the file online and edit simultaneously. Changes are saved automatically.

RESULT:

Excel file was successfully uploaded and shared using SharePoint for collaboration.

OUTPUT IMAGE:


Book1 - Excel
Home
New
Open
Share
Get Add-ins
Info
Save
Save As
Print
Export
Close
Save As
Recent
FabrikamPro
OneDrive - FabrikamPro
jenna@margiestravels.com
Sites - FabrikamPro
jenna@margiestravels.com
Share options
Share
Copy Link
Other locations
This PC
Add a Place
Browse
Jenna Wilcox's OneDrive - FabrikamPro > Documents
Q1 Forecast
Excel Workbook (*.xlsx)
Save
More options...
New Folder
Name
Date modified
Apps
8/12/2024 4:48 PM
Book
2/7/2025 7:47 PM
Q1 Bookings
11/27/2024 11:04 AM